

Carbon and its Compounds (Set B) — MCQ Worksheet

Science • Chapter 4 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Diamond and graphite are:

- (A) Allotropes of carbon
- (B) Isotopes of carbon
- (C) Compounds of carbon and oxygen
- (D) Mixtures

Q2. Fullerenes have:

- (A) Spherical or ellipsoidal carbon clusters
- (B) Layered sheets
- (C) 3D rigid frameworks
- (D) No defined shape

Q3. Which gas is the main constituent of biogas?

- (A) Methane
- (B) Ethane
- (C) Propane
- (D) Butane

Q4. The next member of CH_4 , C_2H_6 , C_3H_8 is:

- (A) C_4H_{10}
- (B) C_4H_8
- (C) C_5H_{10}
- (D) C_3H_6

Q5. Compounds with same molecular formula but different structures are called:

- (A) Isomers
- (B) Isotopes
- (C) Allotropes
- (D) Polymers

Q6. Functional group of ketones is:

- (A) $-\text{OH}$
- (B) $-\text{CHO}$
- (C) $\text{C}=\text{O}$ (in middle)
- (D) $-\text{COOH}$

Q7. IUPAC name of CH_3COCH_3 is:

- (A) Propanone
- (B) Propanal
- (C) Propanoic acid
- (D) Propan-2-ol

Q8. Burning of ethanol gives:

- (A) $\text{CO}_2 + \text{H}_2\text{O}$
- (B) Only H_2O
- (C) Only CO_2
- (D) $\text{CO} + \text{H}_2\text{O}$

Q9. Hydrogenation of vegetable oils converts:

- (A) Unsaturated to saturated fats
- (B) Saturated to unsaturated
- (C) Trans to cis
- (D) All sugar

Q10. Aliphatic carboxylic acid is:

- (A) Ethanoic acid
- (B) Benzoic acid
- (C) Phenol
- (D) Aniline

Q11. Detergents:

- (A) Can work in hard water (calcium/magnesium salts soluble)
- (B) Form scum easily
- (C) Are less effective
- (D) Are biodegradable always

Q12. Vegetable oil left in air develops rancidity due to:

- (A) Oxidation
- (B) Reduction
- (C) Polymerisation
- (D) Hydrolysis

Q13. C_4H_{10} has how many structural isomers?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q14. Which hydrocarbon burns with a sooty flame?

- (A) Methane
- (B) Ethane
- (C) Acetylene (C_2H_2)
- (D) Propane

Q15. Saponification is the reaction of fat/oil with:

- (A) Acid
- (B) Alkali (giving soap and glycerol)
- (C) Salt
- (D) Sugar

Q16. Why does ethanol turn purple-coloured potassium permanganate colourless?

- (A) Ethanol is oxidised to ethanoic acid by KMnO_4
- (B) Ethanol absorbs colour
- (C) It cools KMnO_4
- (D) It dilutes KMnO_4

Q17. Esterification reaction occurs between:

- (A) Carboxylic acid and alcohol (acid catalyst)
- (B) Two acids
- (C) Two alcohols
- (D) Acid and base

Q18. Wash soda is added to detergents because:

- (A) It helps form lather even in hard water
- (B) Improves smell
- (C) Increases volume
- (D) Acts as bleach

Q19. Anhydrous CaCl_2 is used in saponification reaction because:

- (A) It absorbs moisture / facilitates soap precipitation
- (B) It improves smell
- (C) It melts
- (D) It cleans

Q20. Carbon usually forms covalent bonds because:

- (A) It is small with 4 valence electrons; gaining/losing 4 electrons is unfavourable
- (B) It is reactive
- (C) It is metallic
- (D) It is heavy

Q21. The temperature at which the ignition of a substance starts is called:

- (A) Ignition temperature
- (B) Boiling point
- (C) Melting point
- (D) Flash point

Q22. Why is detergent better than soap for washing in hard water?

- (A) Ca/Mg salts of detergent are soluble (no scum)
- (B) Detergent is alkaline
- (C) Detergent is acidic
- (D) Detergent is cheaper

Q23. If hydrocarbon X gives carbon dioxide and water on combustion, then X must contain:

- (A) Carbon and hydrogen only
- (B) Only carbon
- (C) Only hydrogen
- (D) Carbon, hydrogen and oxygen

Q24. Bromine water is decolourised by:

- (A) Unsaturated hydrocarbons (alkenes and alkynes) – addition
- (B) Alkanes
- (C) Pure water
- (D) Methane

Q25. Number of covalent bonds in propane (C_3H_8):

- (A) 8
- (B) 10
- (C) 11
- (D) 12